



James Ruse Agricultural High School

2022 Year 12 TRIAL HSC EXAMINATION

Mathematics Extension 1

General

Instructions

- Reading time – 10 minutes
- Working time – 2 hours
- Write using black pen
- Calculators approved by NESA may be used
- A reference sheet is provided
- For questions in Section II, show relevant mathematical reasoning and/ or calculations

Total marks:
70

Section I – 10 marks (pages 2–5)

- Attempt Questions 1–10
- Allow about 15 minutes for this section

Section II – 60 marks (pages 7–12)

- Attempt Questions 11–14
- Allow about 1 hour and 45 minutes for this section

Section I

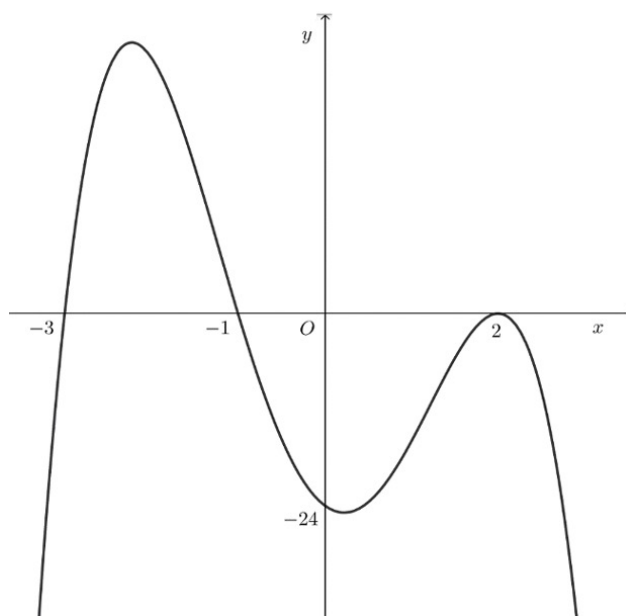
10 marks

Attempt Questions 1 – 10

Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1 – 10.

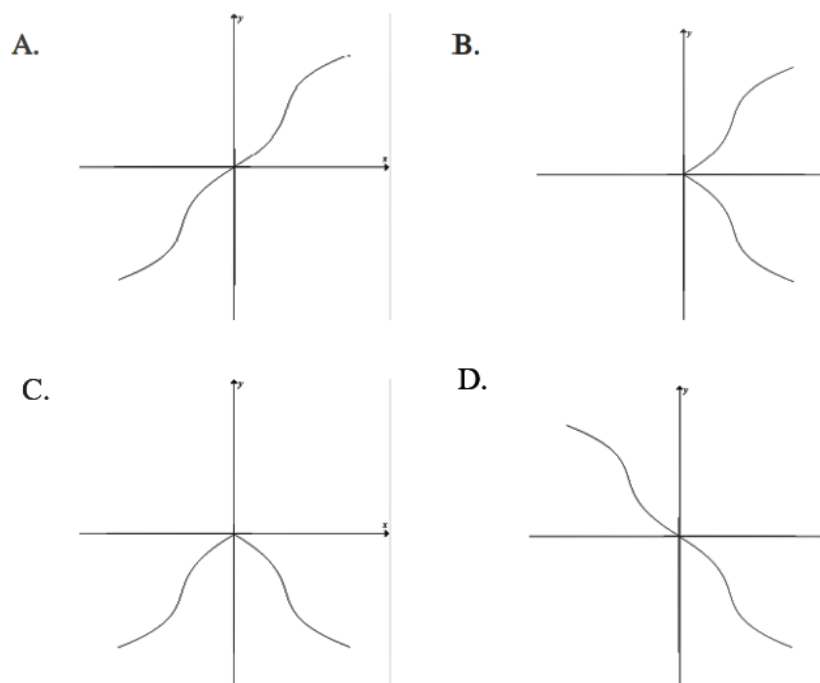
- 1 The diagram below shows the graph of $P(x) = a(x + b)(x + c)(x + d)^2$.



Which of the following is a possible set of values for a, b, c and d ?

- A. $-24, 3, 1, -2$
- B. $-2, 3, 1, -2$
- C. $-2, -1, -3, 2$
- D. $-24, -1, -3, 2$

- 2 $f(x)$ is an odd function and $f(-1) < 0$. A possible graph showing $y = f(-|x|)$ is



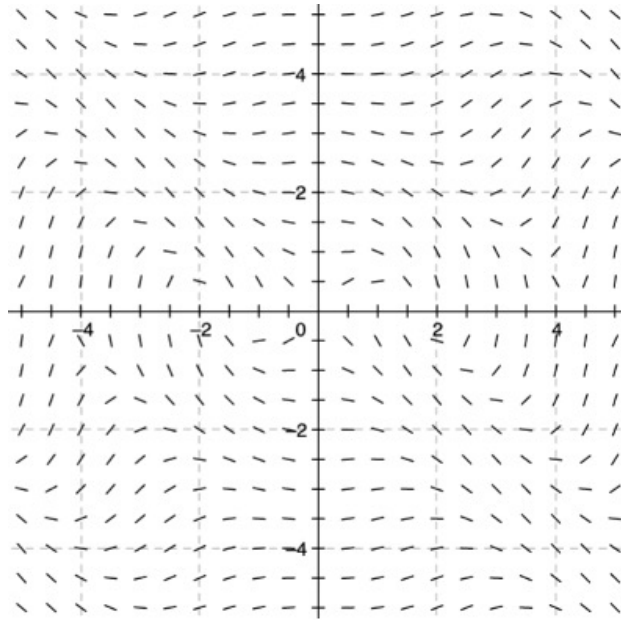
- 3 Which is a correct anti-derivative for $\frac{6}{\sqrt{4-9x^2}}$?

- A. $\sin^{-1}\left(\frac{3x}{2}\right)$
 B. $2 \sin^{-1}\left(\frac{3x}{2}\right)$
 C. $3 \sin^{-1}\left(\frac{3x}{2}\right)$
 D. $6 \sin^{-1}\left(\frac{3x}{2}\right)$

- 4 Consider two hot bodies, A and B, which have temperatures of 120°C and 80°C at $t = 0$ seconds. The temperature of the surrounding is 40°C . The ratio of the rate of cooling of body A to body B at $t = 0$ seconds will be:

- A. 1:2
 B. 2:1
 C. 3:2
 D. 2:3

- 5 Which of the following differential equations yields the direction field below?



- A. $\frac{dy}{dx} = \frac{x \cos(x+y)}{y}$
 B. $\frac{dy}{dx} = \frac{y \cos(x+y)}{x}$
 C. $\frac{dy}{dx} = \frac{x \sin(x+y)}{y}$
 D. $\frac{dy}{dx} = \frac{y \sin(x+y)}{x}$

- 6 A particle has position vector given by $\underline{r}(t) = \cos^{-1}(t) \underline{i} + \sin^{-1}(t) \underline{j}$. The distance travelled by the particle in the interval $[-1, 1]$ is closest to

- A. 6
 B. $\sqrt{3}\pi$
 C. 5
 D. $\sqrt{2}\pi$

7 Given $\cos(3\theta) = 4\cos^3\theta - 3\cos\theta$, which of the following is the simplified form of $\cos^{-1}(4x^3 - 3x)$?

A. $\pi - 3\sin^{-1}x$

B. $3\sin^{-1}x$

C. $3\cos^{-1}x$

D. $\pi - 3\cos^{-1}x$

8 How many integers do not satisfy the inequality $3|x - 1| + x^2 - 7 > 0$?

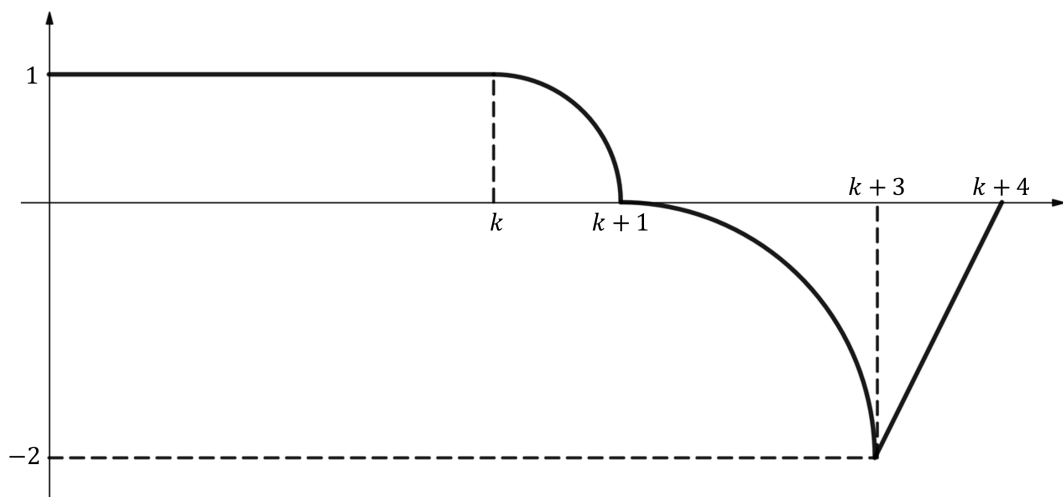
A. 4

B. 3

C. 2

D. 1

9 If $\int_0^{k+4} f(x)dx = 0$, what is the correct value of k ?



A. $6 - \frac{\pi}{4}$

B. $5 - \frac{5\pi}{2}$

C. $6 - \frac{\pi}{2}$

D. $5 - \frac{5\pi}{4}$

10 The graph of $y = \frac{1}{a + bx + 4ax^2}$, where $a > 1$, has only one asymptote when

- A. $-5 < b < 5$
- B. $b \leq -4a$ or $b \geq 4a$
- C. $-1 \leq b \leq 4$
- D. $b < -4a$ or $b > 4a$

Section II

60 marks

Attempt Questions 11 – 14

Allow about 1 hour and 45 minutes for this section

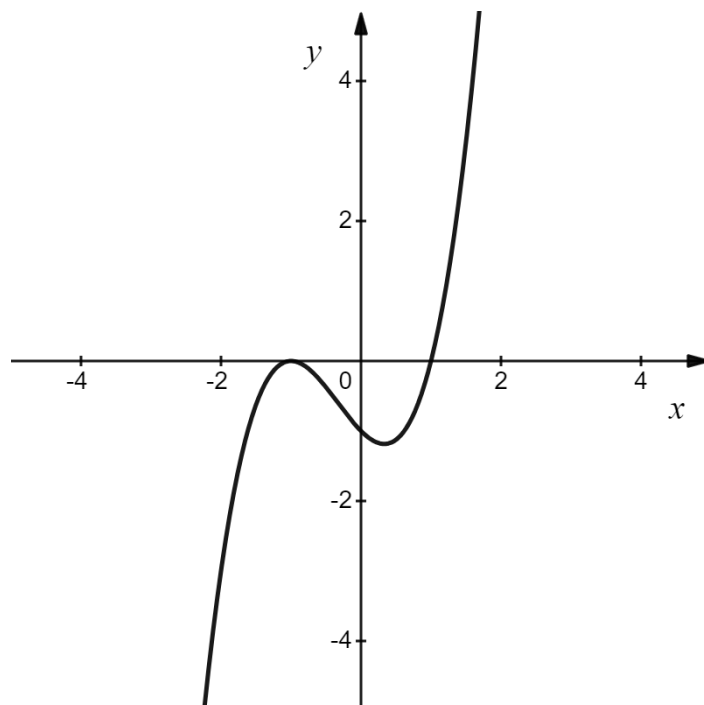
Answer each question on a new page. Extra writing pages are available.

For questions in Section II, your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 marks) Start a new page

- (a) Consider the graph of $f(x) = (x + 1)^2(x - 1)$ given below.

2



Using the axes provided on the reverse of the Multiple Choice answer sheet, sketch the graph of $y^2 = f(x)$, showing all essential features.

- (b) Use the substitution $u = e^{2x}$ to find

3

$$\int \left(e^{e^{2x}}\right)^2 e^{2x} dx$$

Question 11 continues over the page

- (c) Three forces $\vec{F}_1$, $\vec{F}_2$, and $\vec{F}_3$ are represented by vectors $10\hat{i}+20\hat{j}$, $20\hat{i}-30\hat{j}$ and $-6\hat{i}+20\hat{j}$ Newtons respectively.
- (i) Draw a sketch showing the relative magnitude and direction of the three forces. 2
 - (ii) Determine the resultant force, expressed as a vector, and find its magnitude. 2
 - (iii) An additional force $\vec{F}_4 = k\hat{j}$ is now applied so that the resultant of $\vec{F}_1$, $\vec{F}_2$, and $\vec{F}_3$ and $\vec{F}_4$ has magnitude 30N. What is the value of k ? 2
- (d) You are asked to provide an 8-letter password which may only be comprised of the letters A, B and C.
- (i) How many 8-letter passwords are there to choose from? 1
 - (ii) How many of the 8-letter passwords contain only B's and C's? 1
 - (iii) If you decide that your 8-letter password must use each of the letters A, B and C at least once, how many possible passwords are there to choose from? 2

End of Question 11

Question 12 (15 marks) Start a new page

(a) The quartic polynomial $P(x) = x^4 + px^3 + qx^2 + px + 1$ has $x = \alpha$ as a double zero.

(i) Show that α^{-1} is also a zero of $P(x)$. **2**

(ii) Find the fourth root in terms of α . **1**

(iii) Hence show that $p^2 = 4(q - 2)$. **3**

(b) Suppose 3000 people are surveyed on political preference and asked the following question: **3**

“Would you vote for Party A in the next election, yes or no?”

If the actual probability of selecting someone in the affirmative is $p = 0.55$, find, using the normal distribution, the probability that the sample proportion $\hat{p}$ is within 2% of the actual value p .

You may use the z-score table provided.

(c)

(i) Show that the differential equation **2**

$$\frac{dy}{dx} = \frac{2xy}{x^2 - 1}$$

describes a family of parabolas with x -intercepts at $(1, 0)$ and $(-1, 0)$.

(ii) Now consider the differential equation **3**

$$\frac{dy}{dx} = \frac{1 - x^2}{2xy}$$

Find the equation of the curve which satisfies this differential equation and passes through the point $(1, 1)$. Express your answer as a function of x .

(iii) What can be said about the curve in part (ii) in relation to the family of curves in part (i)? **1**

End of Question 12

Question 13 (15 marks) Start a new page

(a) A student is given the chance to win 6^X dollars on two rolls of an unbiased, six-sided die, where X is the number of times a '3' is rolled.

(i) What is the most the student may win? **1**

(ii) How much is the student expected to win? Give your answer correct to two decimal places. **2**

(b) Solve for all $x \in \mathbb{R}$: $\sin x + \cos 2x - \sin 3x = 0$ **3**

(c) Consider the curve $y = 3 \cos^{-1} \left(\frac{x}{2} \right)$.

(i) Sketch the graph of the curve $y = 3 \cos^{-1} \left(\frac{x}{2} \right)$. **2**

(ii) The area bounded by the curve $y = 3 \cos^{-1} \left(\frac{x}{2} \right)$, the y -axis and the lines $y = \frac{\pi}{2}$ and $y = \frac{5\pi}{2}$ is rotated around the y -axis to form an hourglass. Find the volume of the hourglass formed. **4**

(d) Prove by mathematical induction for all positive integers $n \geq 1$ that **3**

$$1^2 \times 2 + 2^2 \times 2^2 + 3^2 \times 2^3 + \dots + n^2 \times 2^n = (n^2 - 2n + 3)2^{n+1} - 6$$

End of Question 13

Question 14 (15 marks) Start a new page

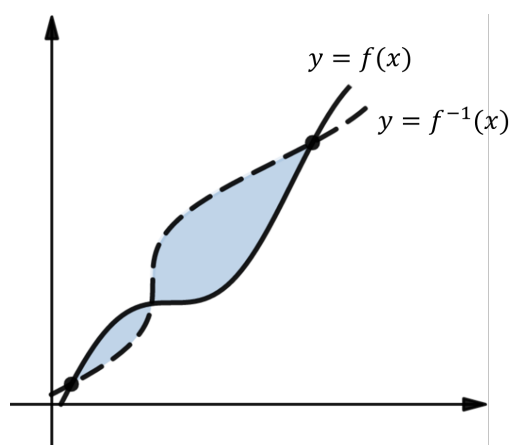
- (a) Let $f(x) = \frac{1}{2}(2x - 1 + 2\sin x)$. The graphs of $y = f(x)$ and $y = f^{-1}(x)$ are shown below.

(i) Find the points of intersection of the two curves.

1

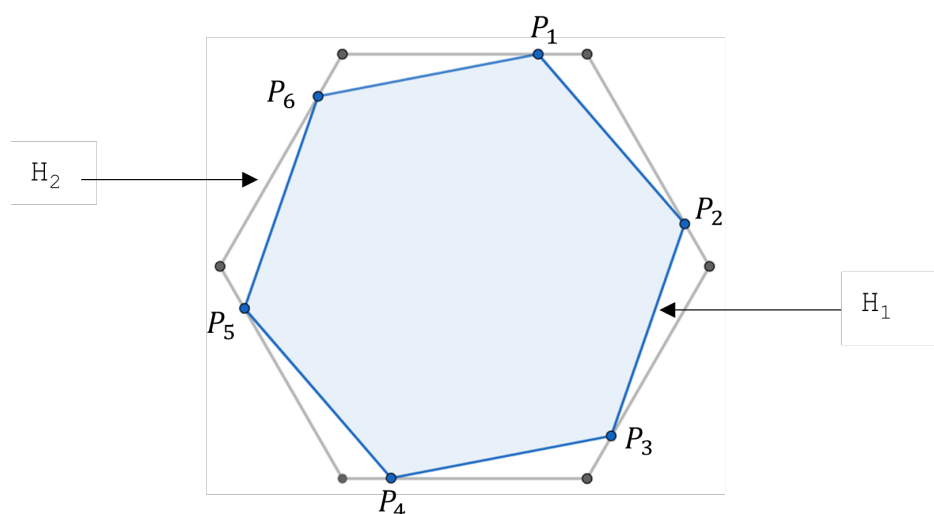
(ii) Find the shaded area enclosed by the two curves.

3



- (b) Six points P_1, P_2, P_3, P_4, P_5 and P_6 form the vertices of a regular hexagon H_1 . An animated program is rotating these six points at a constant speed of 3 cm/s around the perimeter of another regular hexagon, H_2 , which has side length of 20 cm.

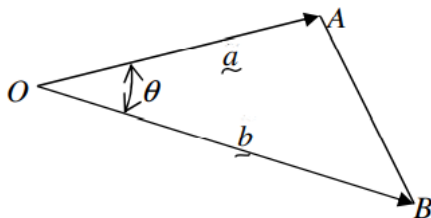
4



Initially, the six vertices of H_1 lie on top of the six vertices of H_2 . Determine the rate of change of the area of H_1 ten seconds after the animation begins.

Question 14 continues over the page

- (c) Triangle OAB is defined by two non-parallel and non-zero vectors $\overrightarrow{OA} = \underline{a}$ and $\overrightarrow{OB} = \underline{b}$. The angle between the vectors $\underline{a}$ and $\underline{b}$ is θ . 3



Express the area of triangle OAB in terms of dot products of $\underline{a}$ and $\underline{b}$.

- (d) Show that $\cos\left(\tan^{-1}\left(\frac{x}{\sqrt{1-x^2}}\right) + \cos^{-1}x\right) = 0$ for all real x , $-1 < x < 1$. 4

You may assume that the range of $\frac{x}{\sqrt{1-x^2}}$ is all real numbers.

End of Question 14

End of Examination.

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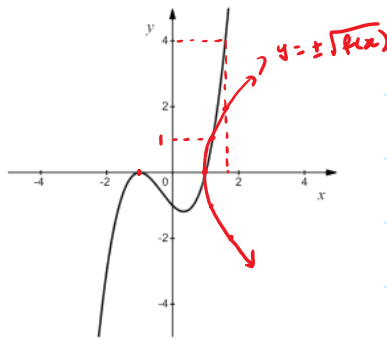
1. B
2. C
3. B
4. B
5. A
6. D
7. C
8. A
9. D
10. C

Question 11

Sunday, 21 August 2022

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a)



1 mark for shape ($\sqrt{f(x)} > f(x)$ for $0 < y < 1$)

1 mark for accuracy (point at $x = -1$ AND either $(2, 3)$ on graph or correct pair for when $y = 2$)

b) $u = e^{2x}$

$$\frac{du}{dx} = 2e^{2x}$$

$$du = 2e^{2x} dx$$

$$\int (e^{2x})^2 e^{2x} dx$$

$$= \frac{1}{2} \int (e^{2x})^2 2e^{2x} dx$$

$$= \frac{1}{2} \int (e^u)^2 du \quad \text{--- 1 mark for correct u substitution into integral}$$

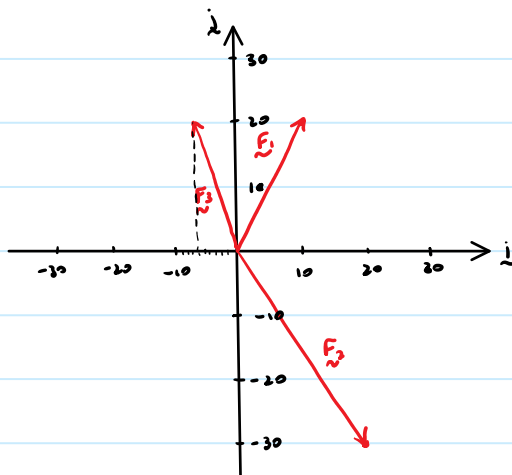
$$= \frac{1}{2} \int e^{2u} du$$

$$= \frac{1}{2} \times \frac{1}{2} e^{2u} \quad \text{--- 1 mark for correct antiderivative in terms of u}$$

$$= \frac{1}{4} e^{2u}$$

$$= \frac{1}{4} e^{2e^{2x}} + C \quad \text{--- 1 mark for re-substituting back } u = e^{2x} \text{ into final answer}$$

c) i)



1 mark for correct quadrants

1 mark for accuracy

Note: All vectors must be on the same diagram
(ie one axis)

If drawn on separate diagrams 1 mark if all
drawn on same scale.

c) ii) $\vec{F}_{net} = 10\hat{i} + 20\hat{j} + 20\hat{i} - 30\hat{j} - 6\hat{i} + 20\hat{j}$

$$= 24\hat{i} + 10\hat{j}$$

$$|\vec{F}_{net}| = \sqrt{24^2 + 10^2}$$

$$= \sqrt{676}$$

$$= 26 \text{ N}$$

① for net force

① for magnitude

iii) $24\hat{i} + 10\hat{j} + k\hat{j}$

$$= 24\hat{i} + (k+10)\hat{j}$$

$$\sqrt{24^2 + (k+10)^2} = 30$$

$$24^2 + (k+10)^2 = 900$$

$$(k+10)^2 = 324$$

$$k+10 = \pm 18$$

$$k = \pm 18 - 10$$

$$= -2, 8$$

① for correct setup

① for correct values of k

d) i) $3^8 = 6561$ ① mark for correct answer

ii) $2^8 = 256$ ① mark for correct answer

Also accepted interpretation of $2^8 - 2 = 254$ which contain only B's & C's but excludes

iii) $3^8 - 3 \times 256 + 3 = 5796$

all B's & all C's

or

$$3^8 - 3 \times 254 = 5796$$

② marks for full correct answer

① for $3^8 - 3 \times 256 = 5793$

MATHEMATICS Extension 1 : Question 1.2...

1

Suggested Solutions

Marks
Awarded

Marker's Comments

$$(i) P(x) = x^4 + px^3 + qx^2 + px + 1$$

$$= 0$$

$$P\left(\frac{1}{x}\right) = \frac{1}{x^4} + \frac{p}{x^3} + \frac{q}{x^2} + \frac{p}{x} + 1$$

$$= \frac{1 + px + qx^2 + px^3 + x^4}{x^4}$$

$$= 0$$

$\therefore P\left(\frac{1}{x}\right)$ is a root

(ii) Product of roots

$$x, x, \frac{1}{x}, \beta$$

$$x^2 \times \frac{1}{x} \times \beta = 1$$

$$\beta = \frac{1}{x}$$

(iii)

sum of the roots

$$x + x + \frac{1}{x} + \frac{1}{x} = -p$$

$$2\left(x + \frac{1}{x}\right) = -p$$

sum of the roots 2 at a time

$$x^2 + x \times \frac{1}{x} + x \times \frac{1}{x} + x \times \frac{1}{x} + 2 \times \frac{1}{x} + \frac{1}{x^2} = q$$

$$q = x^2 + \frac{1}{x^2} + 4$$

1
for $P(x)=0$

• Some students went on incorrect root and made it hard for themselves

1

• students lost $\frac{1}{x^4}$ in their answer.

1

• need to use the term without p or q

• students using other terms struggled to find the root.

1

• some students failed to have negative sign

1

$x^2 + \frac{1}{x^2} + 2$ a common incorrect answer made it difficult to prove correctly

MATHEMATICS Extension 1 : Question...12

Suggested Solutions	Marks Awarded	Marker's Comments
$p^2 = 4\left(x^2 + \frac{1}{x^2} + 2\right)$ $= 4(x-2)$	1	expanding and equating coefficients could gain correct answer.
(b) $0.55 - 0.55 \times 0.02 \leq \hat{p} \leq 0.55 + 0.55 \times 0.02$ $0.539 \leq \hat{p} \leq 0.561$ $\hat{p}^2 = \frac{pq}{n} = \sqrt{0.0000825} = 0.0091$	1	• significant number of students just did $0.55 - 0.02$ and $0.55 + 0.02$ gave 0.9722 as answer
$Z_1 = \frac{0.54 - 0.55}{0.0091} \quad Z_2 = \frac{0.56 - 0.55}{0.0091}$ $= -1.211 \quad = 1.211$	1	
using <u>1</u> <u>normal</u> table $2P(0.8869 - 0.5)$ $= \underline{0.7738}$	1	

MATHEMATICS Extension 1 : Question 13

Suggested Solutions	Marks Awarded	Marker's Comments
c) (i) $\frac{dy}{dx} = \frac{2xy}{x^2-1}$ $\int \frac{dy}{y} = \int \frac{2x}{x^2-1} dx$ $\ln y = \ln x^2-1 + c$ $ y = e^{\ln x^2-1 + c}$ $= e^c x^2-1 $ $y = A(x-1)(x+1)$ which is a family of parabolas with x-intercepts $(-1,0)$ and $(1,0)$ (ii) $\frac{dy}{dx} = \frac{1-x^2}{2xy}$ $\int 2y dy = \int \frac{1-x^2}{x} dx$ $y^2 = \frac{1}{2} \ln x - \frac{x^2}{2} + c$ (1,1) $1 = \ln 1 - \frac{1}{2} + c$ $c = \frac{3}{2}$ $y^2 = \ln x - \frac{x^2}{2} + \frac{3}{2}$ $x=1, y=1$ so $y = \sqrt{\ln x - \frac{x^2}{2} + \frac{3}{2}}$	1 1 1 1 1	well done • substituting (1,0) to evaluate c was meant you were using what you were trying to prove - students write $y = x^2-1 + c$ which does give intercepts $(-1,0)$ and $(1,0)$ • students need to make sure they answer question asked. careless arithmetic errors. needed to write as a function of x $y = \pm \sqrt{\quad}$ is not a function

12

MATHEMATICS Extension 1 : Question.....

Suggested Solutions	Marks Awarded	Marker's Comments
(iii) $\frac{dy}{dx} \times \frac{dy}{dx} = -1$ the curve in part(ii) meets the family of parabolas in part (c) at right angles	1	Poorly done . • inverse functions was common incorrect answer. • Needed to mention right angle to gain mark.

(1)

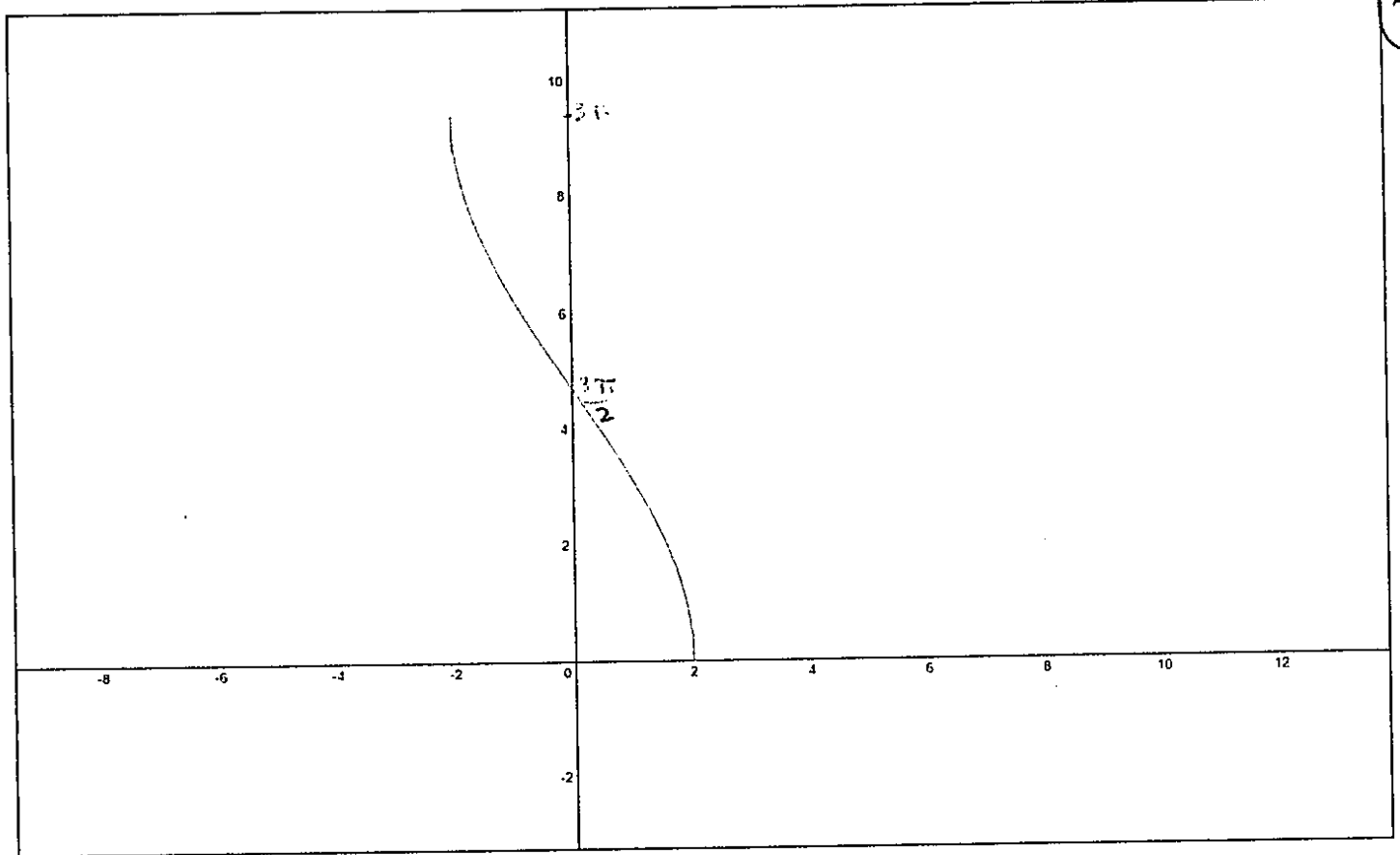
MATHEMATICS Extension 1 : Question 13...

Suggested Solutions	Marks Awarded	Marker's Comments
a) i) $6^2 = 36$ (ii) $P = \frac{1}{6} E(6^x)$ $\therefore$ $= 6^0 \binom{2}{0} \left(\frac{1}{6}\right)^0 \left(\frac{5}{6}\right)^2 + 6^1 \binom{2}{1} \left(\frac{1}{6}\right)^1 \left(\frac{5}{6}\right)^1 + 6^2 \binom{2}{2} \left(\frac{1}{6}\right)^2 \left(\frac{5}{6}\right)^0$ $= \frac{1}{36} (25 + 60 + 36)$ $= \$3.36$	1 1	well done 1 for binomial 2 for correct answer • poorly done most students used 0, 1, 2 not $6^0, 6^1, 6^2$ $\\$1.82$ common incorrect answer. • student who showed binomial gained 1 mark.
b) $\sin x + \cos 2x - \sin 3x = 0$ $\sin x - \sin 3x = 2 \cos 2x \sin(-x)$ $\therefore \cos 2x - 2 \cos 2x \sin x = 0$ $\cos 2x (1 - \sin x) = 0$ $\cos 2x = 0 \text{ or } \sin x = \frac{1}{2}$	1 1	1 mark for correct manipulation of trig result 2 marks for gaining correct equation

2

MATHEMATICS Extension 1: Question 13...

Suggested Solutions	Marks Awarded	Marker's Comments
b) $2x = (2k+1)\frac{\pi}{2} \therefore x = (2k+1)\frac{\pi}{4}$ $x = k\pi + (-1)^k \cdot \frac{\pi}{6} \quad k \in \mathbb{Z}$ c) see graph. (vi) $x = 2\cos\left(\frac{y}{3}\right)$ $x^2 = 4\cos^2\frac{y}{3}$ $V = \pi \int_{\frac{\pi}{2}}^{\frac{3\pi}{2}} x^2 dy + \pi \int_{\frac{3\pi}{2}}^{\frac{5\pi}{2}} x^2 dy$ $= 2\pi \int_{\frac{\pi}{2}}^{\frac{3\pi}{2}} 4\cos^2\frac{y}{3} dy$ $= 2\pi \int_{\frac{\pi}{2}}^{\frac{3\pi}{2}} 2(1 + \cos\frac{2y}{3}) dy$ $= 4\pi \left[\frac{2y}{2} + \frac{3}{2} \sin\frac{2y}{3} \right]_{\frac{\pi}{2}}^{\frac{3\pi}{2}}$ $= 4\pi \left(\frac{3\pi}{2} + \frac{3}{2} \sin\pi - \frac{\pi}{2} - \frac{3}{2} \sin\frac{\pi}{3} \right)$ $= 4\pi \left[\pi - \frac{3\sqrt{3}}{4} \right]$	1 1 1 for converting $\cos^2\frac{y}{3}$ 1 for integration 1	needed to generate all solutions needed $k \in \mathbb{Z}$ if write solution, needed negative as well. Solutions with incorrect limits needed to evaluate $\sin\frac{\pi}{3}$ to gain mark.



1 mark for correct shape

- Many students made it look like a horizontal point of inflexion and did not gain mark

$$\sim y = 3 \cos^{-1}\left(\frac{x}{2}\right)$$

1 mark for correct position

- Students used poor scales which distorted graph because they were different on each axis.

MATHEMATICS Extension 1 : Question 13...

Suggested Solutions	Marks Awarded	Marker's Comments
(d) $n=1$ $LHS = 1 \times 2$ $RHS = (1-2+3)2-6$ $= 2$ $= 2$ $\therefore LHS = RHS$ so true for $n=1$ Prove true for $n=k$ $1^2 \times 2 + 2^2 \times 2^2 + \dots + k^2 \times 2^k = (k^2 - 2k + 3) \times 2^{k+1} - 6$ Assume true for $n=k+1$ $1^2 \times 2 + \dots + k^2 \times 2^k + (k+1)^2 \times 2^{k+1}$ $= ((k+1)^2 - 2(k+1) + 3) \times 2^{k+2} - 6$ $\therefore 1^2 \times 2 + \dots + k^2 \times 2^k + (k+1)^2 \times 2^{k+1}$ $= (k^2 - 2k + 3)2^{k+1} - 6 + (k+1)^2 2^{k+1}$ (using assumption) $= 2^{k+1}(k^2 - 2k + 3 + k^2 + 2k + 1) - 6$ $= 2^{k+1}(2k^2 + 4) - 6$ $= 2^{k+1}(k^2 + 2) - 6$	1 1	• must write LHS and RHS not as one long statement. well done. need to make it clear when you are using assumption

MATHEMATICS X1 : Question.....

Suggested Solutions	Marks Awarded	Marker's Comments
Now $RHS = (k^2 + 2k + 1 - 2k - 2 + 3)2^{k+2} - 6$ $= (k^2 + 2)2^{k+2} - 6$ $= LHS$ $\therefore$ true by process of mathematical induction for $n \geq 1$.		

Question 14

Sunday, 21 August 2022 9:27 PM

a) i) Point of intersection of $f(x)$ & $f^{-1}(x)$ is the same as point of intersection of $f(x)$ & x

$$\frac{1}{2}(2x-1+2\sin x) = x$$

$$2x-1+2\sin x = 2x$$

$$2\sin x = 1$$

$$\sin x = \frac{1}{2}$$

$$x = \frac{\pi}{6} + 2k\pi, \frac{5\pi}{6} + 2k\pi \quad k \in \mathbb{Z}$$

or

$$x = \frac{\pi}{6}, \frac{5\pi}{6}, \frac{13\pi}{6}$$

① mark for correct general solution or correct first 3 solutions

$$\text{ii) } A = 2 \int_{\frac{\pi}{6}}^{\frac{5\pi}{6}} f(x) - x \, dx + 2 \int_{\frac{5\pi}{6}}^{\frac{13\pi}{6}} x - f(x) \, dx$$

$$= 2 \int_{\frac{\pi}{6}}^{\frac{5\pi}{6}} \frac{1}{2}(2x-1+2\sin x) - x \, dx + 2 \int_{\frac{5\pi}{6}}^{\frac{13\pi}{6}} x - \frac{1}{2}(2x-1+2\sin x) \, dx \quad \text{① for correct setup}$$

$$= \int_{\frac{\pi}{6}}^{\frac{5\pi}{6}} 2x-1+2\sin x - 2x \, dx + \int_{\frac{5\pi}{6}}^{\frac{13\pi}{6}} 2x - (2x-1+2\sin x) \, dx$$

$$= \int_{\frac{\pi}{6}}^{\frac{5\pi}{6}} 2\sin x - 1 \, dx + \int_{\frac{5\pi}{6}}^{\frac{13\pi}{6}} 1 - 2\sin x \, dx$$

$$= \left[-2\cos x - x \right]_{\frac{\pi}{6}}^{\frac{5\pi}{6}} + \left[x + 2\cos x \right]_{\frac{5\pi}{6}}^{\frac{13\pi}{6}}$$

$$= \left(-2\cos \frac{5\pi}{6} - \frac{5\pi}{6} \right) - \left(-2\cos \frac{\pi}{6} - \frac{\pi}{6} \right) + \left(\frac{13\pi}{6} + 2\cos \frac{13\pi}{6} \right) - \left(\frac{5\pi}{6} + 2\cos \frac{5\pi}{6} \right)$$

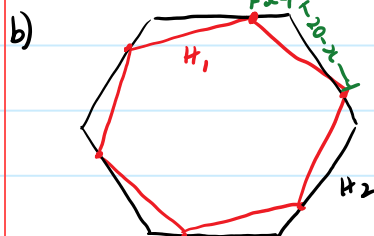
$$= -2x - \frac{\sqrt{3}}{2} - \frac{5\pi}{6} + 2x \frac{\sqrt{3}}{2} + \frac{\pi}{6} + \frac{13\pi}{6} + 2x \frac{\sqrt{3}}{2} - \frac{5\pi}{6} - 2x - \frac{\sqrt{3}}{2}$$

$$= 13 - \frac{5\pi}{6} + 13 + \frac{\pi}{6} + \frac{13\pi}{6} + 13 - \frac{5\pi}{6} + 13$$

$$= 13 + \frac{2\pi}{3}$$

① for correct antiderivative

① for evaluating incorrectly



let x be the distance each vertex on H_1 travels after t seconds

$$\frac{dx}{dt} = 3 \text{ cm/s}$$

let A = area of H_1

$$\frac{dA}{dt} = \frac{dA}{dx} \times \frac{dx}{dt}$$

let l be the side length of H_1

$$\begin{aligned} l^2 &= x^2 + (20-x)^2 - 2x(20-x) \cos 120^\circ \\ &= x^2 + 400 - 40x + x^2 - 2x(20-x)(-\frac{1}{2}) \\ &= x^2 + 400 - 40x + x^2 + 20x - x^2 \\ &= x^2 - 20x + 400 \end{aligned}$$

$$A = 6 \times \frac{1}{2} \times l \times l \times \sin 60^\circ$$

$$= \frac{3\sqrt{3}}{2} l^2$$

$$= \frac{3\sqrt{3}}{2} (x^2 - 20x + 400)$$

$$\frac{dA}{dx} = \frac{3\sqrt{3}}{2} (2x - 20)$$

$$= 3\sqrt{3} (x - 10)$$

$$\frac{dA}{dt} = 3\sqrt{3} (x - 10) \times 3$$

$$= 9\sqrt{3} (x - 10)$$

After 10s, the point has travelled $3 \times 10 = 30 \text{ cm}$, $30 - 20 = 10$ so it is 10cm from the vertex

ie need to sub in $x = 10$

$$\frac{dA}{dt} = 9\sqrt{3} (10 - 10)$$

$$= 0 \text{ cm}^2/\text{s}$$

① for finding an expression for the side length of H_1

① for finding an expression for the area of H_1

① for finding correct value of x to sub in

① for finding the correct expression for $\frac{dA}{dt}$ and evaluating correctly when $x = 10$.

c) $a \cdot b = |a||b| \cos \theta$ — ①

$$A = \frac{1}{2} |a||b| \sin \theta$$

$2A = |a||b| \sin \theta$ — ②

①² + ②² $(a \cdot b)^2 + (2A)^2 = |a|^2 |b|^2 \cos^2 \theta + |a|^2 |b|^2 \sin^2 \theta$

$$= |a|^2 |b|^2 (\cos^2 \theta + \sin^2 \theta)$$

$$= |a|^2 |b|^2$$

$$4A^2 = |a|^2 |b|^2 - (a \cdot b)^2$$

$$2A = \sqrt{|a|^2 |b|^2 - (a \cdot b)^2} \quad (A > 0)$$

$$A = \frac{1}{2} \sqrt{|a|^2 |b|^2 - (a \cdot b)^2}$$

$$= \frac{1}{2} \sqrt{(a \cdot a)(b \cdot b) - (a \cdot b)^2}$$

① mark for using Pythagorean identity

① mark for simplifying to this expression

① mark for final expression only in dot products

$$A = \frac{1}{2} a \cdot b \tan \theta$$

$$A = \frac{1}{2} |a||b| \sin \left(\cos^{-1} \left(\frac{a \cdot b}{|a||b|} \right) \right)$$

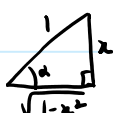
} both got ① mark

d) let $\alpha = \tan^{-1} \frac{x}{\sqrt{1-x^2}}$, $\beta = \cos^{-1} x$

$$\tan \alpha = \frac{x}{\sqrt{1-x^2}}$$

$$x = \cos \beta$$

When $0 < x < 1$ and $0 < \alpha < \frac{\pi}{2}$

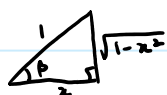


$$\cos \alpha = \sqrt{1-x^2}$$

$$\sin \alpha = x$$

① mark for drawing triangles correctly & fully labeled.

When $0 < x < 1$ and $0 < \beta < \frac{\pi}{2}$



$$\cos \beta = x$$

$$\sin \beta = \sqrt{1-x^2}$$

① mark for considering all cases, not just the acute cases

When $-1 < x < 0$ and $-\frac{\pi}{2} < \alpha < 0$



$$\cos \alpha = \sqrt{1-x^2}$$

$$\sin \alpha = -x$$

When $-1 < x < 0$ and $\frac{\pi}{2} < \beta < \pi$



$$\cos \beta = -x, \sin \beta = \sqrt{1-x^2}$$

$$\cos \left(\tan^{-1} \left(\frac{x}{\sqrt{1-x^2}} \right) \right) + \cos^{-1} x = 0 \quad \text{for } x \in (-1, 1)$$

$$\cos \left(\tan^{-1} \left(\frac{x}{\sqrt{1-x^2}} \right) \right) \cos(\cos^{-1} x) - \sin \left(\tan^{-1} \left(\frac{x}{\sqrt{1-x^2}} \right) \right) \sin(\cos^{-1} x)$$

$$(\sqrt{1-x^2})x - x\sqrt{1-x^2} = 0 \quad \text{for all cases}$$

① mark for expanding using compound angles

① mark for showing that the expression = 0